

What a Neural Jastrow Factor Can and Cannot Learn: A Controlled Variational Monte Carlo Study of the Beryllium Atom

Francesco Agus¹

¹*Student ID: 265962*

(Dated: September 13, 2026)

We report a Variational Monte Carlo (VMC) study of the ground state of the beryllium atom ($Z = 4$) designed as a controlled experiment on the role of the Jastrow correlation factor. Four trial wave functions sharing an *identical*, jointly re-optimised Slater determinant are compared: the bare determinant, an analytic Padé Jastrow enforcing the exact Kato cusp conditions, a neural Jastrow in which a shared multi-layer perceptron acts on each interelectronic distance and must learn the cusps by itself, and a hybrid in which the network acts on a cusp-safe coordinate $t(r) = r^2/(1+r)$ so that the imposed cusps are provably untouched. A fifth ansatz extends the determinant to a minimal complete active space (CAS) built from the $2s^2$ and $2p^2$ configurations and adds a three-body electron-electron-nucleus neural Jastrow. All wave functions are optimised by Stochastic Reconfiguration; the Laplacian entering the local energy is evaluated exactly by forward-over-reverse automatic differentiation in JAX. Within the shared-determinant series the variational energy saturates at $-14.6331(12) E_h$ (63.7% of the correlation energy), whereas a single additional variational parameter acting on the determinant reaches $-14.66662(28) E_h$ (99.2%). We show that a 340-parameter neural Jastrow lowers the variance of the local energy by only 4% relative to a five-parameter analytic form, that the network learns the antiparallel-spin cusp to within 0.7% but misses the parallel-spin one by 81%, and that imposing the exact cusps yields no measurable energy gain for this system. These observations are traced to an analytical fact: the Jastrow factor is strictly positive and therefore cannot move the nodal surface, while the parallel-spin channel is a sloppy direction to which the energy is nearly insensitive because antisymmetry has already depleted the corresponding region of configuration space.

I. INTRODUCTION

1. The many-electron problem

Solving the stationary non-relativistic Schrödinger equation for interacting electrons remains a central problem of computational physics. For an N -electron atom with a clamped nucleus of charge Z , the Hamiltonian reads, in Hartree atomic units ($\hbar = m_e = e = 1, 4\pi\epsilon_0 = 1$),

$$\hat{H} = -\frac{1}{2} \sum_{i=1}^N \nabla_i^2 - \sum_{i=1}^N \frac{Z}{r_i} + \sum_{i<j}^N \frac{1}{r_{ij}}, \quad (1)$$

where $r_i = |\mathbf{r}_i|$ and $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. The pairwise repulsion $\sum_{i<j} r_{ij}^{-1}$ couples all coordinates, so $\hat{H}\Psi = E\Psi$ cannot be reduced to independent single-particle equations. Direct discretisation of the $3N$ -dimensional configuration space is ruled out by the exponential growth of the required grid, and exact diagonalisation by the corresponding growth of the Hilbert space dimension.

Variational Monte Carlo sidesteps the dimensionality problem rather than solving it. One posits a parametrised trial wave function $\Psi_\theta(\mathbf{R})$ with $\mathbf{R} \in \mathbb{R}^{3N}$, and estimates the Rayleigh quotient

$$E(\theta) = \frac{\langle \Psi_\theta | \hat{H} | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} = \mathbb{E}_{\mathbf{R} \sim |\Psi_\theta|^2} [E_L(\mathbf{R})], \quad (2)$$

with the local energy defined as

$$E_L = \frac{\hat{H}\Psi_\theta}{\Psi_\theta}, \quad (3)$$

by sampling configurations from $|\Psi_\theta|^2$ with a Markov chain. The integral is never evaluated: only the ratio $E_L(\mathbf{R})$ is, pointwise. By the variational principle $E(\theta) \geq E_0$ for every θ , the trial wave function parameters, so the method returns a rigorous upper bound whose quality is set entirely by the expressive power of the chosen ansatz. The design of Ψ_θ is therefore the physics of the calculation.

2. Correlation energy, dynamic and static

The natural reference point is the Hartree-Fock (HF) approximation, in which Ψ is a single antisymmetrised product of optimised single-particle orbitals. HF is variational and captures shell structure and exchange exactly, but it replaces the instantaneous electron-electron interaction by a mean field. The energy it misses defines the correlation energy,

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}, \quad (4)$$

which for beryllium amounts to $-0.0944 E_h$ against $E_{\text{HF}} = -14.5730 E_h$ and $E_{\text{exact}} = -14.6674 E_h$.

It is essential for what follows to separate two contributions of very different physical origin. *Dynamic* correlation is the short-range mutual avoidance of electrons: the cusp-like depletion of $|\Psi|^2$ as $r_{ij} \rightarrow 0$, which is a local, largely universal feature. *Static* correlation arises when several electronic configurations are nearly degenerate and the exact ground state is an essential superposition of them, so that no single determinant can be a good starting point. The two are attacked by different tools.

Dynamic correlation is the natural domain of a multiplicative correlation factor; static correlation is not, for a reason developed below.

Beryllium is an unusually instructive test case precisely because it exhibits both. With only four electrons it is small enough to be treated without approximation beyond the ansatz itself, yet its $2s^2$ configuration is nearly degenerate with $2p^2$, and roughly half of E_{corr} is static in origin. A method that captures only dynamic correlation will visibly stall part way, and the point at which it stalls is quantitatively predictable.

3. The Slater-Jastrow ansatz and its structural limit

The standard correlated ansatz multiplies a determinant by a positive correlation factor,

$$\Psi_\theta(\mathbf{R}) = D(\mathbf{R}) e^{J(\mathbf{R})}, \quad (5)$$

where D is antisymmetric and J is a symmetric function of the electron coordinates. Traditionally J is a compact analytic form, such as a Padé expression in the interelectronic distances, containing a handful of parameters. A recent and rapidly growing line of work replaces such hand-designed forms by neural networks, exploiting their status as universal approximators to obtain far more flexible correlation factors.

The structural observation that organises this paper is the following. Since $e^J > 0$ everywhere, the Jastrow factor cannot change the sign of Ψ_θ : the nodal surface, the set where $\Psi_\theta = 0$, is fixed entirely by D . A Jastrow factor can redistribute amplitude, but it cannot move a single node. Consequently, no amount of expressive power added to J can carry a Slater-Jastrow wave function past the best energy attainable with the nodal structure of D . If the missing correlation is static in origin, and therefore encoded in the nodes, adding parameters to the Jastrow is the wrong parametrization.

This is well known in principle. It is rarely demonstrated, because published comparisons typically vary the determinant and the correlation factor at the same time, so that the two effects cannot be disentangled.

4. This work

We design the comparison so that the two effects *are* disentangled. Four trial wave functions are constructed which share an identical determinant, parametrised by three variational parameters and independently re-optimised in each case, and which differ *only* in the correlation factor:

- (A) the bare determinant, $J = 0$;
- (B) an analytic Padé Jastrow, in which the exact electron-electron cusp conditions hold by construction;

- (C) a neural Jastrow, in which a single multi-layer perceptron with shared weights acts on each interelectronic distance, and which is free to represent a cusp but must discover it by optimisation;

- (D) a hybrid, in which the Padé factor imposes the cusps and the network acts on the cusp-safe coordinate $t(r) = r^2/(1+r)$, for which $t'(0) = 0$, so that the network provably cannot contribute any slope at coalescence.

Any energy difference within this series is attributable to the Jastrow factor alone. A fifth ansatz, denoted (D+), then breaks the shared-determinant constraint: it promotes D to a minimal complete active space spanned by the $1s^2 2s^2$ and $1s^2 2p_k^2$ configurations with a single variational mixing coefficient c_p , and pairs it with a three-body electron-electron-nucleus neural Jastrow. Model (D+) does not belong to the controlled series and is analysed separately; its purpose is to demonstrate what the series cannot reach.

All five wave functions are optimised by Stochastic Reconfiguration, the stochastic realisation of natural gradient descent, which is essential here because orbital parameters and network weights differ by orders of magnitude in scale. The Metropolis sampler uses single-electron moves to decorrelate the tightly bound core and the diffuse valence electrons at comparable rates.

Beyond the energies themselves, the comparison yields three results that we believe are of general interest and that a single model calculation cannot expose. First, the neural Jastrow of model (C) learns the antiparallel-spin cusp to within 0.7% of its exact value while missing the parallel-spin cusp by 81%; imposing both exactly, as in model (D), changes the energy by less than one standard error. Second, the parallel-spin channel is a *sloppy* direction: models (B) and (D) converge to parallel-spin correlation functions that are qualitatively opposite, yet agree in energy, because antisymmetry has already depleted the region of configuration space where two like-spin electrons approach one another. Third, in the absence of imposed cusps the local energy develops heavy tails severe enough that the per-epoch energy estimator occasionally falls below bounds it cannot physically violate.

5. Organisation

Section II specifies the determinant, the orbitals and the way in which the electron-nucleus and electron-electron cusp conditions are imposed, and defines the five correlation factors. Section III details the evaluation of the local energy, the Monte Carlo sampler, the Stochastic Reconfiguration update with its damping and trust-region prescriptions, and the estimation of statistical errors by blocking. Section IV presents the controlled series, the learned cusps, the optimised correlation functions, and the multi-determinant ansatz. Section V draws the gen-

eral conclusions, and Section VI states the limitations of the study and the natural extensions.

II. THEORY AND WAVE FUNCTION ANSATZ

1. Hamiltonian, spin assignment and antisymmetry

For the neutral beryllium atom ($Z = 4$, $N = 4$) Eq. (1) specialises to

$$\hat{H} = -\frac{1}{2} \sum_{i=0}^3 \nabla_i^2 - \sum_{i=0}^3 \frac{4}{r_i} + \sum_{0 \leq i < j \leq 3} \frac{1}{r_{ij}}, \quad (6)$$

with $\mathbf{R} = (\mathbf{r}_0, \dots, \mathbf{r}_3) \in \mathbb{R}^{12}$.

In the absence of spin-orbit coupling the Hamiltonian does not act on spin, and we may work in a fixed $S_z = 0$ spin assignment: electrons 0,1 carry spin up and electrons 2,3 spin down. Antisymmetry is then enforced separately within each spin channel by a product of two 2×2 determinants,

$$D(\mathbf{R}) = \det M_{\uparrow}(\mathbf{r}_0, \mathbf{r}_1) \det M_{\downarrow}(\mathbf{r}_2, \mathbf{r}_3), \quad (7)$$

where, for a pair of spatial orbitals (ϕ_a, ϕ_b) ,

$$M_{\uparrow} = \begin{pmatrix} \phi_a(\mathbf{r}_0) & \phi_b(\mathbf{r}_0) \\ \phi_a(\mathbf{r}_1) & \phi_b(\mathbf{r}_1) \end{pmatrix}, \quad M_{\downarrow} = \begin{pmatrix} \phi_a(\mathbf{r}_2) & \phi_b(\mathbf{r}_2) \\ \phi_a(\mathbf{r}_3) & \phi_b(\mathbf{r}_3) \end{pmatrix}. \quad (8)$$

Exchanging two electrons of the same spin exchanges two rows of the corresponding matrix and flips the sign of D , as required. There are exactly six distinct pairs (i, j) , of which two are parallel-spin, (0,1) and (2,3), and four antiparallel.

The six pair distances r_{ij} and the four radial coordinates r_i are computed with a regularised norm, $r = (\sum_{\alpha} x_{\alpha}^2 + \varepsilon)^{1/2}$ with $\varepsilon = 10^{-12}$, so that the gradient of the norm remains finite at coincidence; this affects the arithmetic only within 10^{-6} bohr of a singularity.

2. Orbitals and the electron-nucleus cusp

Whenever an electron approaches the nucleus, the potential $-Z/r_i$ diverges. A finite local energy requires the kinetic term to produce a compensating divergence, which for an s -type orbital imposes the electron-nucleus cusp condition

$$\left. \frac{\phi'(r)}{\phi(r)} \right|_{r \rightarrow 0} = -Z. \quad (9)$$

We impose Eq. (9) exactly rather than leaving it to the optimiser. For the $1s$ orbital this fixes the exponent: taking $\phi_{1s} \propto e^{-\alpha_{1s}r}$ gives $\phi'/\phi = -\alpha_{1s}$, so $\alpha_{1s} = Z = 4$ is held *fixed* throughout. Promoting α_{1s} to a variational parameter would lower the energy slightly while destroying the cusp and inflating the variance of the local energy; we do not do so.

For the $2s$ orbital we adopt the form $\phi \propto (r - c) e^{-\alpha_{2s}r}$, whose radial node at $r = c$ is what distinguishes it from the $1s$. Expanding about the origin,

$$\phi(0) = -c, \quad \phi'(r) = e^{-\alpha_{2s}r} [1 - \alpha_{2s}(r - c)] \quad (10)$$

this implies

$$\phi'(0) = 1 + \alpha_{2s}c, \quad (11)$$

so that Eq. (9) reads $(1 + \alpha_{2s}c)/(-c) = -Z$ and determines the node position uniquely,

$$c = \frac{1}{Z - \alpha_{2s}}. \quad (12)$$

Thus c is not an independent parameter: it is recomputed from α_{2s} at every evaluation of the wave function, which keeps the cusp exact as α_{2s} varies during the optimisation.

To grant the determinant additional radial freedom without spoiling Eq. (9), we multiply each orbital by a factor built on the coordinate

$$t(r) = \frac{r^2}{1 + r}, \quad t(0) = 0, \quad t'(0) = 0. \quad (13)$$

Because $t'(0) = 0$, a factor $(1 + pt)$ has unit value and zero slope at the origin, so the logarithmic derivative at $r = 0$ is unchanged. The orbitals used in models (A)–(D) are therefore

$$\phi_{1s}(r) = e^{-Zr} (1 + p_1 t(r)), \quad (14)$$

$$\phi_{2s}(r) = (r - c) e^{-\alpha_{2s}r} (1 + p_2 t(r)), \quad (15)$$

with three variational parameters $\{\alpha_{2s}, p_1, p_2\}$, initialised at (0.98, 0, 0) and softly bounded to prevent the appearance of spurious nodes.

It is common to Gram-Schmidt the $2s$ orbital against the $1s$ before building the determinant. A determinant is invariant under the addition of a multiple of one column to another, so for any λ

$$\det(\phi_{1s}, \phi_{2s} - \lambda \phi_{1s}) = \det(\phi_{1s}, \phi_{2s}), \quad (16)$$

identically and for every configuration $\mathbf{R}$. The orthogonalisation coefficient, whatever its value, cancels exactly from the wave function. We therefore omit it. The redundant construction introduces overlap integrals that must be recomputed whenever α_{2s} changes, at no benefit.

3. Multi-configuration determinant

Model (D+) replaces the single configuration by a minimal complete active space (CAS). In addition to ϕ_{1s} and ϕ_{2s} , which for this model are used without the $(1 + pt)$ factors, we introduce three $2p$ orbitals

$$\phi_{2p_k}(\mathbf{r}) = x_k e^{-\alpha_{2p}r}, \quad k \in \{x, y, z\}, \quad (17)$$

which vanish linearly at the nucleus and therefore carry no electron-nucleus cusp condition, and which are orthogonal

to ϕ_{1s} and ϕ_{2s} by angular symmetry. Writing $D[\chi]$ for the product of determinants (7) built from the orbital pair (ϕ_{1s}, χ) , the multi-determinant wave function is

$$D_{\text{CAS}}(\mathbf{R}) = D[\phi_{2s}] + c_p \sum_{k \in \{x, y, z\}} D[\phi_{2p_k}], \quad (18)$$

with a *single* variational mixing coefficient c_p , initialised at zero so that the optimisation starts from the single-determinant wave function.

The three Cartesian p^2 configurations must enter with equal coefficients for the state to remain an $L = 0$ singlet. Model (D+) thus adds exactly one parameter to the nodal structure and three to the orbitals (α_{2s} , α_{2p} , c_p ; the p_i factors are dropped).

Crucially, c_p is the only parameter in the entire study that moves the nodal surface. Everything else either rescales orbitals or enters through a strictly positive factor.

4. Electron-electron cusp conditions

When two electrons coalesce, $r_{ij} \rightarrow 0$, the repulsion $1/r_{ij}$ diverges and the same cancellation argument applies to the relative coordinate. Kato's condition for the spherically averaged wave function reads

$$\left. \frac{1}{\Psi} \frac{\partial \Psi}{\partial r_{ij}} \right|_{r_{ij} \rightarrow 0} = a_{ij}, \quad a_{ij} = \begin{cases} \frac{1}{4} & \text{parallel spins,} \\ \frac{1}{2} & \text{antiparallel spins.} \end{cases} \quad (19)$$

The two values differ because for parallel spins the determinant already vanishes linearly at coalescence, so half of the required cancellation is supplied by antisymmetry and the correlation factor need only provide the rest. This asymmetry turns out to control much of what the neural network does and does not learn.

Since D contributes a fixed amount, imposing Eq. (19) on $\Psi = De^J$ amounts to a condition on the Jastrow exponent alone, $\partial J / \partial r_{ij} \rightarrow a_{ij}$ as $r_{ij} \rightarrow 0$.

5. The five correlation factors

All models share the structure $\Psi_\theta = De^J$ and differ only in J . We define the spin label $s_{ij} = +1$ for parallel and $s_{ij} = -1$ for antiparallel pairs.

a. (A) Bare determinant. $J \equiv 0$. No cusp condition is satisfied at electron-electron coalescence, and the local energy diverges there.

b. (B) Padé Jastrow.

$$J_{\text{P}}(\mathbf{R}) = \sum_{i < j} \frac{a_{ij} r_{ij}}{1 + b_{ij} r_{ij}}, \quad (20)$$

with $b_{ij} = b_{\parallel}$ or $b_{\perp}$ according to s_{ij} . Differentiating at the origin gives $\partial J_{\text{P}} / \partial r_{ij}|_0 = a_{ij}$ independently of b_{ij} : the cusps are imposed exactly by construction, and the two free parameters control only the range of the correlation hole. They are bounded to $b \in [0.05, 10]$; the significance of the upper bound is discussed in Sec. IV.

c. (C) Neural Jastrow.

$$J_{\text{NN}}(\mathbf{R}) = \sum_{i < j} f_\theta \left(\frac{r_{ij}}{1 + r_{ij}}, s_{ij} \right), \quad (21)$$

where f_θ is a single multi-layer perceptron with weights shared across all six pairs. Weight sharing enforces exact permutation symmetry among equivalent pairs. The input $x = r/(1+r)$ maps $[0, \infty)$ onto $[0, 1)$, which conditions the network well, and satisfies $dx/dr = 1$ at $r = 0$: the network is therefore *able* to represent a cusp of any slope, but nothing forces it to adopt the correct one. The spin flag is required because a single scalar channel could not reproduce two different cusp values.

d. (D) Hybrid.

$$J_{\text{H}}(\mathbf{R}) = J_{\text{P}}(\mathbf{R}) + \sum_{i < j} f_\theta \left(\frac{t(r_{ij})}{1 + t(r_{ij})}, s_{ij} \right), \quad (22)$$

with $t(r)$ as in Eq. (13). Since $t'(0) = 0$, we have that $\partial(f_\theta \circ x)/\partial r|_{r=0} = 0$ for any weights: the network cannot contribute slope at coalescence, and the cusps remain exactly those imposed by the Padé term. The network capacity is thereby restricted to the region where correlation is not already captured by the exact condition. This provides a numerical check with a known answer, exploited in Sec. IV.

e. (D+) Three-body neural Jastrow. For the multi-determinant model the network is promoted to a three-body electron-electron-nucleus form,

$$J_3(\mathbf{R}) = J_{\text{P}}(\mathbf{R}) + \sum_{i < j} g_\theta(t_i + t_j, t_i t_j, t_{ij}, s_{ij}), \quad (23)$$

where $t_i \equiv t(r_i)$ and $t_{ij} \equiv t(r_{ij})$. The features are symmetric under $i \leftrightarrow j$, which is mandatory: an asymmetric pair function would make $|\Psi|^2$ depend on the labelling of identical particles and destroy the permutation symmetry of the density. Building every feature on t guarantees that neither the electron-electron nor the electron-nucleus cusps are affected, since $t'(0) = 0$. The network output is multiplied by an envelope $\exp[-\alpha_{\text{env}}(t_i + t_j + t_{ij})]$ with $\alpha_{\text{env}} = 0.3$.

6. Network architecture and numerical choices

Both f_θ and g_θ are multi-layer perceptrons with two hidden layers of 16 units, denoted by $\mathbf{h}^{(i)}$. Denoting the input feature vector by $\mathbf{u}$,

$$\begin{aligned} \mathbf{h}^{(1)} &= \sigma(W^{(1)}\mathbf{u} + \mathbf{b}^{(1)}), \\ \mathbf{h}^{(2)} &= \sigma(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}), \\ f_\theta &= W^{(3)}\mathbf{h}^{(2)} + b^{(3)}, \end{aligned} \quad (24)$$

where $W^{(i)}$ are the weights and with $\sigma(x) = x/(1 + e^{-x})$ the SiLU activation. The choice of activation is dictated

TABLE I. The five variational ansätze. Models (A)–(D) share an identical three-parameter determinant and differ only in the correlation factor; model (D+) extends the determinant to a minimal CAS and is analysed separately. The last column indicates whether the electron-electron cusp conditions, Eq. (19), are imposed by construction, learned by the network, or absent.

Model	Jastrow	Param.	e-e cusps
(A)	none	3	absent
(B)	Padé	5	imposed
(C)	network, r_{ij}	340	learned
(D)	Padé + network, $t(r_{ij})$	342	imposed
(D+)	Padé + three-body network	374	imposed

by the physics: the local energy requires the Laplacian of $\ln|\Psi|$, so the network must be at least twice differentiable with non-vanishing second derivative. A rectified linear unit has $\partial^2\sigma/\partial x^2 = 0$ almost everywhere and would yield a wave function with no curvature contribution from the network.

Two implementation details deserve mention. First, the output layer is initialised from a normal distribution of width 10^{-2} rather than from zero. A zero-initialised output layer makes the network output identically zero, which is attractive because the ansatz then starts exactly at the analytic baseline; but it also makes the gradients with respect to *all* hidden weights vanish, since $\partial f/\partial W^{(1)} \propto W^{(3)} = 0$. The optimisation would then explore only the handful of parameters in the last layer. Second, the parameter dtype must be set explicitly to `float64`: the framework default is single precision, which is silently retained even when double precision is enabled globally, and which degrades the conditioning of the Stochastic Reconfiguration.

Table I summarises the five ansätze.

III. COMPUTATIONAL METHODOLOGY

1. Local energy in logarithmic form

The wave function is represented throughout by its logarithm, $L(\mathbf{R}) = \ln|\Psi_\theta(\mathbf{R})|$, which avoids overflow and makes the Jastrow factor additive. Writing $\Psi = \pm e^L$ and using $\nabla\Psi = \Psi\nabla L$,

$$\frac{\nabla^2\Psi}{\Psi} = \nabla^2 L + |\nabla L|^2, \quad (25)$$

so that the local energy takes the form actually implemented,

$$E_L(\mathbf{R}) = -\frac{1}{2}\left(\nabla^2 L + |\nabla L|^2\right) - Z \sum_i \frac{1}{r_i} + \sum_{i<j} \frac{1}{r_{ij}}, \quad (26)$$

with the Laplacian and gradient taken over all 12 Cartesian coordinates. The sign of Ψ , which changes across nodes in the multi-determinant model, never enters: only

$|\Psi|$ ratios are needed for sampling and only L for the energy.

2. Metropolis sampling with single-electron moves

Configurations are drawn from $P_\theta \propto |\Psi_\theta|^2$ by a Metropolis random walk. We use $N_w = 4000$ independent walkers evolved in parallel, initialised from Gaussians of width 0.4 for the two core electrons and 1.8 for the two valence electrons.

Rather than displacing all four electrons simultaneously, each move displaces a single electron,

$$\mathbf{r}_i \rightarrow \mathbf{r}_i + \delta\boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_3), \quad (27)$$

and is accepted with probability $\min(1, |\Psi_\theta(\mathbf{R}')/\Psi_\theta(\mathbf{R})|^2)$, the proposal being symmetric. A *sweep* consists of four such moves, one per electron. Single-electron moves are preferable here because the two shells live on very different length scales: the 1s electrons are localised within ~ 0.25 bohr while the valence electrons extend beyond 2 bohr, so an all-electron move must use a step short enough for the core, which then decorrelates the valence electrons very slowly.

The step δ is adapted between epochs by a multiplicative rule targeting an acceptance ratio of 55%; in all production runs it stabilised near $\delta \approx 0.37$ with acceptance between 55% and 58%. Each optimisation epoch is preceded by ten sweeps, and the chains are thermalised by 300 sweeps before optimisation begins.

3. Stochastic Reconfiguration

Define the logarithmic derivative operators, or scores,

$$O_k(\mathbf{R}) = \frac{\partial}{\partial\theta_k} \ln|\Psi_\theta(\mathbf{R})|, \quad (28)$$

computed for each walker giving an $N_w \times N_\theta$ matrix. The exact energy gradient is the correlation between the scores and the local energy,

$$\frac{\partial E}{\partial\theta_k} = 2 \left\langle (O_k - \langle O_k \rangle) (E_L - \langle E_L \rangle) \right\rangle, \quad (29)$$

which follows from differentiating Eq. (2), using $\nabla_\theta P_\theta = 2P_\theta(O - \langle O \rangle)$ and the fact that the term involving $\nabla_\theta E_L$ vanishes by hermiticity of $\hat{H}$. Equation (29) embodies the zero-variance principle: as Ψ_θ approaches an eigenstate, E_L becomes constant, the second factor vanishes pointwise, and both the gradient and its statistical error go to zero.

Plain gradient descent on Eq. (29) is poorly suited to this problem, because the parameters are heterogeneous: a change of 0.01 in the mixing coefficient c_p alters the wave function far more than a change of 0.01 in a network

weight. Stochastic Reconfiguration replaces the Euclidean metric by the Fisher metric of the sampled distribution,

$$S_{kl} = \left\langle (O_k - \langle O_k \rangle)(O_l - \langle O_l \rangle) \right\rangle, \quad (30)$$

and takes the preconditioned step $\delta = S^{-1}\mathbf{g}$, where $\mathbf{g}$ denotes the gradient (29).

Three regularisations are required in practice.

a. Damping. S is estimated from a finite sample and is close to singular along directions to which the energy is insensitive. We damp it relative to its own diagonal,

$$S \rightarrow S + \gamma \text{diag}(S) + \epsilon \mathbb{I}, \quad (31)$$

with γ in the range 10^{-3} – 5×10^{-3} and $\epsilon = 10^{-8}$. Diagonal-relative damping is essential: a uniform $\gamma \mathbb{I}$ would regularise orbital and network directions by wildly different relative amounts.

b. Trust region. The step is bounded in norm *after* multiplication by the learning rate,

$$\Delta\theta = \eta \delta \cdot \min\left(1, \frac{\Delta_{\max}}{\|\eta \delta\|}\right), \quad (32)$$

so that $\|\Delta\theta\| \leq \Delta_{\max}$. Bounding δ before applying η , as is sometimes done, makes the maximum step length $\eta \Delta_{\max}$, which for typical values is orders of magnitude smaller than intended and stalls the optimisation entirely.

c. Tail clipping. Near nodal surfaces E_L develops heavy tails, and the sample mean entering Eq. (29) is not robust. We therefore clip the local energies to a window of five mean absolute deviations about the median,

$$E_L \rightarrow \text{clip}(E_L, m - 5\Delta, m + 5\Delta), \quad (33)$$

with $m = \text{median}(E_L)$ and $\Delta = \langle |E_L - m| \rangle$. The clipping enters the gradient only; all reported energies are computed from unclipped samples.

The learning rate and the trust region are annealed together by a cosine schedule with a 5% floor, $\eta(n) = \eta_0 [0.05 + 0.475(1 + \cos(\pi n/N))]$. The floor is deliberate: a schedule reaching exactly zero produces a vanishing update norm at the final epoch for reasons unrelated to convergence, which is easily mistaken for a converged optimisation.

4. Energy estimation and error bars

After optimisation the parameters are frozen and the energy is accumulated over $N_B = 40$ blocks, each separated by 150 sweeps of the frozen wave function. The long separation is necessary: with 30 sweeps between blocks the residual autocorrelation is large enough to underestimate the error bar by roughly a factor of three.

The uncertainty is obtained by a blocking analysis. Starting from the sequence of block means, we repeatedly average adjacent pairs and record the naive standard error at each level of coarse-graining; the reported error is the

maximum over levels, which is the plateau value when one exists and a conservative bound otherwise.

As an independent diagnostic we evaluate the kinetic and potential energies separately and form the ratio $\langle V \rangle / \langle T \rangle$, which equals -2 for any exact eigenstate of a Coulomb Hamiltonian. It must be interpreted with care in the present setting. E is stationary at the minimum and therefore accurate to second order in the error of Ψ_θ , whereas $\langle T \rangle$ and $\langle V \rangle$ separately are accurate only to first order. The ratio measures the missing freedom, not the accuracy of the energy.

5. Computational details

The code is written in JAX with Flax, in double precision throughout, and is just-in-time compiled and vectorised over walkers. All runs use $N_w = 4000$ walkers, ten sweeps per epoch, and a fixed random seed. The number of epochs is 300 for model (A), 600 for (B), 1500 for (C) and (D), and 2500–3500 for (D+); the smaller models converge in far fewer epochs than the neural ones. On a single laptop CPU the wall-clock cost is about 10 s for model (A), 21 s for (B), 150–180 s for (C) and (D), and 460 s for the longest (D+) run. Repeating the shared-determinant series with an unchanged seed reproduced all energies to the fifth decimal.

The sample-to-parameter ratio is 10.7 for the largest model, comfortably above the rule of thumb of four to ten samples per parameter required for the estimate of S in Eq. (30) to be usable. This consideration, rather than expressive power, sets the network width at 16 hidden units.

IV. RESULTS

1. The shared-determinant series

Table II collects the converged energies of all five ansätze and Fig. 1 shows the optimisation trajectories of the shared-determinant series together with the variance of the local energy.

The bare determinant, model (A), converges to $-14.5696 E_h$, which lies 3.4 mHartree *above* the Hartree-Fock limit and therefore corresponds to a formally negative recovered correlation fraction. This is not a failure of the optimisation: our determinant carries only three radial parameters, whereas the Hartree-Fock limit corresponds to fully relaxed numerical orbitals. Model (A) is a variational approximation to Hartree-Fock, and its energy is bounded from below by it. The comparison is nonetheless meaningful, since all four models in the series carry the same three parameters.

Adding the analytic Padé factor lowers the energy by 53.6 mHartree to $-14.6232 E_h$, recovering 53.2% of the correlation energy with two additional parameters. Replacing the Padé form by a 337-weight neural network

TABLE II. Converged variational energies for the beryllium atom in Hartree atomic units. Uncertainties are blocking estimates over 40 blocks separated by 150 sweeps. The recovered fraction of correlation energy is computed with respect to $E_{\text{HF}} = -14.5730 E_h$ and $E_{\text{exact}} = -14.6674 E_h$. Models (A)–(D) share an identical determinant; model (D+) does not.

Model	Par.	$E (E_h)$	Error	$\text{Var}(E_L)$	Corr.
(A)	3	-14.56959	0.00448	3.2082	-3.6%
(B)	5	-14.62320	0.00113	0.2562	53.2%
(C)	340	-14.63292	0.00112	0.2460	63.5%
(D)	342	-14.63312	0.00119	0.2497	63.7%
(D+)	374	-14.66662	0.00028	0.0156	99.2%
HF limit		-14.5730			0%
FN-DMC, HF nodes		-14.6570			89%
Exact		-14.6674			100%

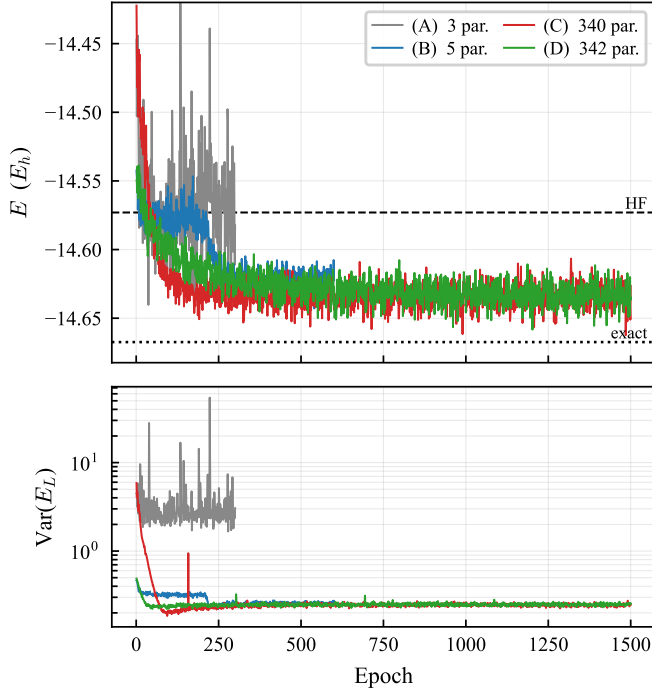


FIG. 1. Optimisation trajectories of the four shared-determinant ansatz (upper panel) and the corresponding variance of the local energy (lower panel, logarithmic scale). The horizontal lines mark the Hartree-Fock limit and the exact non-relativistic ground-state energy. The per-epoch energies are sample means over 4000 walkers and are considerably noisier than the final blocked estimates of Table II.

gains a further 9.7 mHartree, reaching 63.5%. Adding the exact cusps back on top of the network, model (D), changes the energy by 0.20 mHartree, which against a combined uncertainty of 1.6 mHartree is 0.12 standard deviations: models (C) and (D) are statistically indistinguishable.

The variance of the local energy behaves differently from the energy and is more informative. It falls by a factor of 12.5 between models (A) and (B), from 3.21 to 0.256, then essentially stops: 0.246 for (C) and 0.250 for

TABLE III. Electron-electron cusp slopes $\partial J/\partial r_{ij}$ evaluated at $r_{ij} = 0$ for the optimised correlation factors, compared with the exact values of Eq. (19). Model (C) must discover the cusps by optimisation; models (B) and (D) satisfy them by construction.

Pair	Exact	(B)	(C)	(D)
parallel	0.2500	0.2500	0.4529	0.2500
antiparallel	0.5000	0.5000	0.4964	0.5000

(D). The single dominant contribution to the variance is therefore the treatment of the electron-electron coalescence points, which the two-parameter Padé form already handles exactly. The 335 additional parameters of model (C) reduce the variance by 4%.

2. Learned versus imposed cusps

Because the correlation factor is available in closed form for every model, the electron-electron cusp condition can be tested directly by differentiating the optimised Jastrow at the origin. Table III reports the result.

Model (D) returns the exact values to four decimals, confirming numerically the argument of Sec. II 5: a network acting on the coordinate $t(r) = r^2/(1+r)$ cannot contribute slope at coalescence, whatever its weights. This is a check with a known answer and validates the construction.

The behaviour of model (C) is more interesting. The network recovers the antiparallel cusp to within 0.7% of its exact value, but assigns the parallel pairs a slope of 0.4529 instead of 0.2500, an error of 81%. In effect the network has learned a single cusp value close to 1/2 and applied it to both channels, despite having a spin flag available to distinguish them.

The asymmetry has a physical origin. For parallel spins the determinant already vanishes at coalescence, so the region $r_{ij} \rightarrow 0$ carries very little probability weight for like-spin pairs and contributes correspondingly little to the energy. Of the six pairs, four are antiparallel and only two parallel, and those two are further suppressed by the node. The optimiser therefore receives almost no gradient signal from the parallel channel and has no incentive to fix it.

3. Optimised correlation functions

Figure 2 shows the optimised Jastrow exponents, shifted to $J(0) = 0$, together with the distribution of interelectronic distances actually sampled by the walkers.

Three features deserve comment.

First, in model (B) the parallel-spin range parameter converges to $b_{\parallel} = 10.000$, exactly the upper bound imposed in Eq. (20). Since $a r/(1 + b r) \rightarrow a/b$ as b grows, a large $b_{\parallel}$ makes the parallel Jastrow a constant, and constants are irrelevant in the exponent: the optimiser is

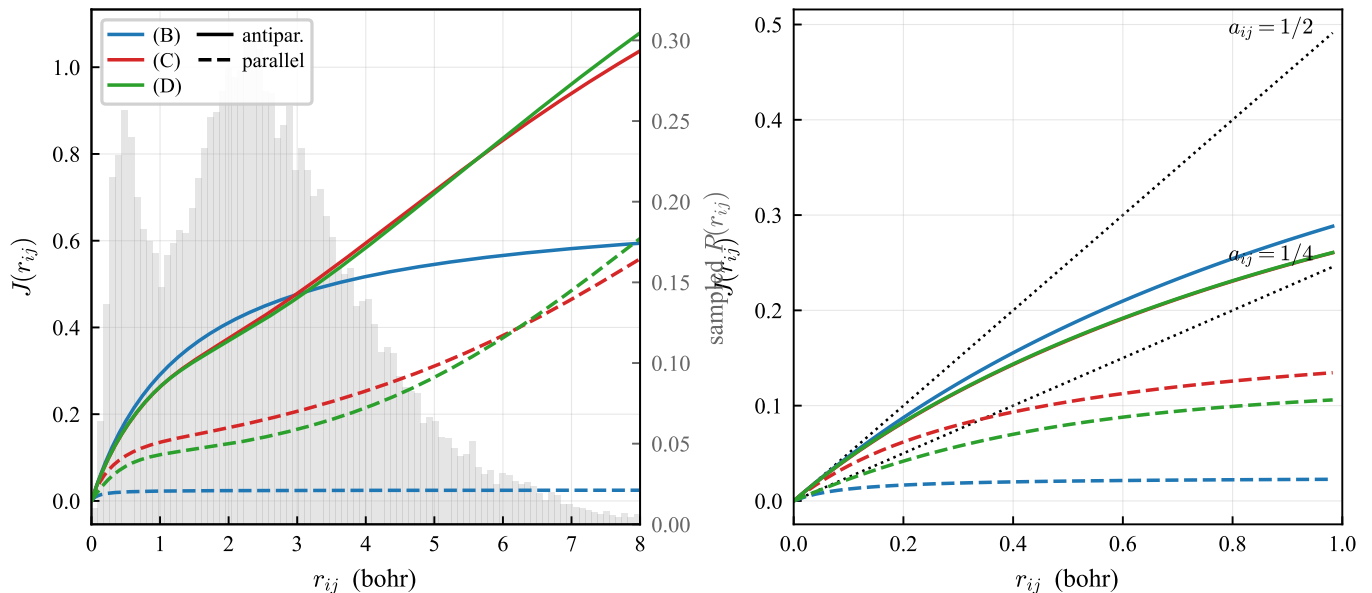


FIG. 2. Optimised Jastrow exponents for models (B), (C) and (D), shifted so that $J(0) = 0$. Solid lines denote antiparallel pairs and dashed lines parallel pairs. The shaded histogram is the distribution of interelectronic distances sampled at convergence, plotted against the right-hand axis; beyond $r_{ij} \approx 5$ bohr the curves are unconstrained extrapolations rather than optimised functions. Right: the coalescence region, with the exact cusp tangents of slope $1/4$ and $1/2$ shown as dotted lines.

switching the parallel-spin correlation factor off. We verified this interpretation by repeating the optimisation with the bound raised to 100, whereupon $b_{\parallel}$ again ran to the boundary and the total energy degraded to $-14.5787 E_h$, close to model (A). The bound is therefore not a cosmetic constraint but a stabiliser, and the value $b_{\parallel} = 10$ must be read as “at the boundary” rather than as an optimised parameter. By contrast, the antiparallel range converges cleanly to $b_{\perp} = 0.7165$.

Second, in model (D) the same parameter converges to $b_{\parallel} = 0.2834$, well inside the allowed interval, with $b_{\perp} = 0.4490$. The presence of the network changes the optimisation landscape enough that the parallel Padé term is no longer driven away. That two models can disagree so completely about the parallel-spin correlation function while agreeing on the energy to within 0.12σ is itself the main evidence for the argument developed in Sec. V 3.

Third, the neural correlation function of model (C) does not saturate: it grows almost linearly at large r_{ij} , with slopes of approximately 0.088 for antiparallel and 0.07 for parallel pairs. A physical two-body correlation factor should approach a constant, since correlation dies out at large separation. Summing over the three partners of a receding electron, one antiparallel and two parallel or the reverse, gives a Jastrow contribution $J \approx 0.25 r$, so the wave function acquires a factor $e^{+0.25r}$ that partially cancels the orbital decay $e^{-\alpha_{2s}r}$. The network is thus performing a one-body task, effectively renormalising the orbital exponent. The optimised parameters confirm this: model (C) converges to $\alpha_{2s} = 1.3135$, against 1.1698 for model (B), the difference compensating the exponential

growth of the correlation factor. This redundancy between network and orbital parameters is a nearly degenerate direction in parameter space, and is one reason why model (C) requires more than twice as many epochs as model (B) to converge.

Finally, the sampled distribution of r_{ij} is concentrated between 0.5 and 4 bohr. The region where the three models differ most dramatically, $r_{ij} \gtrsim 5$ bohr, is essentially never visited, and the apparent divergence between the curves there reflects the absence of data rather than a physical disagreement.

4. Reliability of the energy estimator

At epoch 200 the per-epoch estimator of model (A) reads $-14.6220 E_h$. This value is 48.9 mHartree below the Hartree-Fock limit and is therefore *impossible*: no single determinant, however parametrised, can lie below $-14.5730 E_h$. The final blocked estimate for the same model is $-14.5696 E_h$, correctly above the limit.

The explanation is the variance. Model (A) has no electron-electron cusp, so E_L diverges as r_{ij}^{-1} at coalescence and its distribution has heavy tails, with $\text{Var}(E_L) = 3.21$ against 0.25 for the correlated models. The naive standard error of a 4000-walker mean is then $0.024 E_h$, and the excursion is roughly two of these; with heavy tails the effective error is larger still, since the sample mean converges slowly and the tail is undersampled. The per-epoch estimator is simply not reliable at the millihartree level for this ansatz, whereas the blocked estimate, built from decorrelated blocks, is.

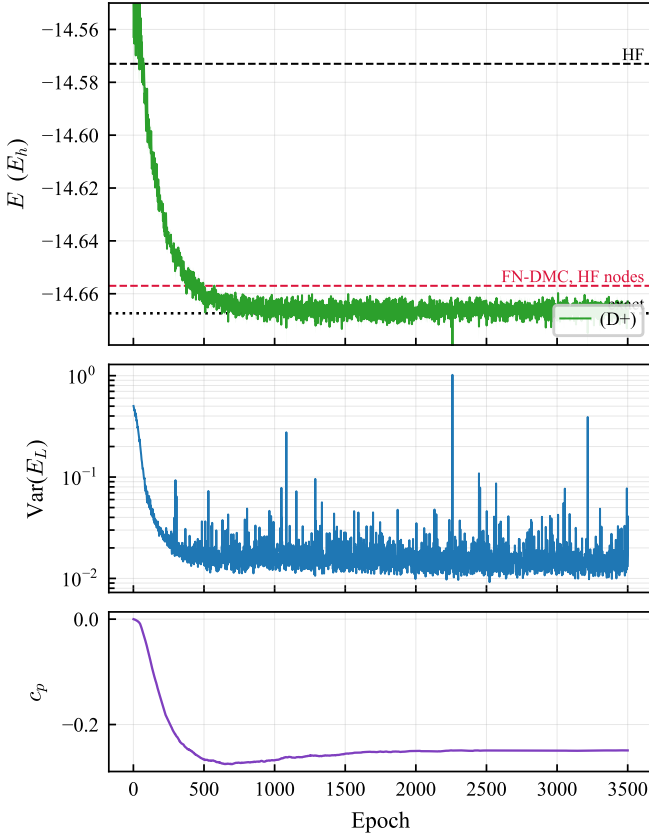


FIG. 3. Optimisation of the multi-determinant ansatz (D+) over 3500 epochs. The line marked FN-DMC with HF nodes is a bound for wave functions built on a single Hartree-Fock determinant; it does not constrain model (D+), whose nodal surface is modified by c_p .

The same phenomenon appears in model (D+), where several per-epoch values fall below the exact ground-state energy, the lowest being $-14.6707 E_h$ at epoch 2200. Against the converged value and the per-epoch scatter this is a 1.9σ fluctuation, and among the thirty-six sampled epochs one or two such excursions are expected. The blocked estimate, $-14.66662(28) E_h$, lies safely above $-14.6674 E_h$. No violation of the variational principle occurs at any point.

5. Beyond the nodes: the multi-determinant ansatz

Model (D+) breaks the shared-determinant constraint by admitting the $2p^2$ configurations through the single mixing coefficient c_p of Eq. (18). Its optimisation is shown in Fig. 3 and its converged parameters in Table IV.

The longer run reaches $E = -14.66662(28) E_h$, recovering 99.2% of the correlation energy and lying 0.78 mHartree above the exact value. The variance of the local energy falls to 0.0156, sixteen times smaller than in the shared-determinant series, consistent with the zero-variance principle.

TABLE IV. Converged parameters of the multi-determinant ansatz for two runs of different length. The energies agree within 0.7 combined standard deviations and the variances are indistinguishable, whereas $\langle T \rangle$ differs by $0.29 E_h$.

	2500 epochs	3500 epochs
$E (E_h)$	$-14.66631(35)$	$-14.66662(28)$
$\text{Var}(E_L)$	0.0157	0.0156
Correlation recovered	98.8%	99.2%
c_p	-0.2556	-0.2491
α_{2s}	1.0330	1.0750
α_{2p}	1.0174	1.0524
$b_{\parallel}$	0.5850	0.4907
$b_{\perp}$	0.5310	0.5510
$\langle T \rangle$	14.7125	14.4204
$\langle V \rangle / \langle T \rangle$	-1.9968	-2.0171

The trajectory makes the mechanism explicit. The energy decreases in lockstep with the growth of $|c_p|$: at epoch 100, with $c_p = -0.056$, the energy is $-14.593 E_h$, comparable to the best shared-determinant model; by epoch 500, with $c_p = -0.266$, it has reached $-14.661 E_h$; thereafter c_p plateaus and so does the energy. The converged value $c_p \approx -0.25$ is the magnitude expected from the near-degeneracy of the $2s^2$ and $2p^2$ configurations in beryllium.

It should be emphasised that the fixed-node bound of $-14.6570 E_h$ quoted in Table II applies to wave functions built on Hartree-Fock nodes. Once $c_p \neq 0$ the nodal surface is no longer that of a single determinant and the relevant bound moves down to approximately $-14.667 E_h$. A variational energy below $-14.6570 E_h$ is therefore entirely legitimate here, and is the direct signature that the nodes have changed.

Two caveats accompany this result. First, model (D+) is not converged in the strict sense: the norm of the Stochastic Reconfiguration update remained pinned at the trust-region ceiling for the entire run, decreasing only because the ceiling itself was annealed. The physically meaningful parameters c_p and α_{2s} have plateaued, but the network weights had not stopped moving. The reported energy is an upper bound that a longer run might improve slightly.

Second, comparing the two runs of Table IV shows that the extra 1000 epochs lowered the energy by 0.31 mHartree, less than the combined uncertainty, while leaving the variance unchanged at 0.0156. Since energy and variance must improve together as Ψ_θ approaches an eigenstate, a gain in energy unaccompanied by a gain in variance indicates that the additional epochs did not improve the wave function. The residual limitation is the ansatz, not the optimisation.

The virial ratio illustrates the same point from another angle. It changes from -1.9968 to -2.0171 between the two runs, while $\langle T \rangle$ shifts by $0.29 E_h$ and the total energy by $0.0003 E_h$: kinetic and potential energies move by a thousand times more than their sum, in opposite directions. The optimiser is travelling along a direction of

radial rescaling to which the energy is nearly flat but the individual expectation values are not. This is precisely the behaviour anticipated in Sec. III, and it means the virial ratio should not be read as a measure of accuracy.

V. DISCUSSION

1. Expressive power in the Jastrow cannot substitute for nodal freedom

Going from the analytic Padé factor to a 337-weight neural network gains 9.7 mHartree. Adding a single variational parameter acting on the determinant gains 33.7 mHartree. The first intervention redistributes amplitude within a fixed nodal structure, the second changes the nodal structure itself.

This is the expected consequence of $e^J > 0$. All four shared-determinant models are confined to the same nodal surface, and the energy they saturate at, $-14.633 E_h$, is far above the exact value not because the correlation factor is insufficiently flexible but because the static correlation associated with the $2s^2-2p^2$ near-degeneracy is encoded in the nodes, where no correlation factor can reach.

The practical implication is that expressive power should be allocated where the physics requires it. For a system with significant static correlation, enlarging the neural Jastrow is close to the least effective use of additional parameters.

2. Imposing exact conditions is free; learning them is not

Model (D) demonstrates that the exact cusp conditions can be imposed while retaining full neural flexibility elsewhere, at the cost of a single change of variable. The measured cusp slopes of Table III are exact to four decimals by construction.

Imposing exact conditions is not pointless, but, the condition model (C) failed to learn was the one the energy is insensitive to. The guarantee provided by model (D) is structural and holds for any system; the observation that it does not matter is specific to beryllium, where only two of six pairs are parallel and the relevant region is suppressed by antisymmetry. In a system with a larger fraction of like-spin pairs, or in a calculation where the variance rather than the energy is the target, the same construction could become useful.

3. Sloppy directions dominate the optimisation

Several independent observations point to the same structural feature of the problem. The network of model (C) misses the parallel-spin cusp by 81%. Models (B) and (D) converge to parallel-spin correlation functions that are qualitatively opposite, one switched off entirely and one of comparable magnitude to the antiparallel channel,

yet their energies agree. And within model (D+), radial rescaling moves $\langle T \rangle$ by three orders of magnitude more than it moves E .

These are all manifestations of directions in parameter space along which the energy is nearly flat. Such directions make parameters non-identifiable, and they produce the appearance of disagreement between models that are in fact equally good. They also explain why Stochastic Reconfiguration is worth here. It measures displacement in the space of wave functions rather than in the space of parameters, and therefore does not spend large steps on directions that change the parameters a great deal and the physics very little.

4. Variance is not a secondary observable

Throughout this study the variance of the local energy carried information the energy alone did not. It revealed that the neural Jastrow of model (C), with 67 times as many parameters, improves on the analytic form by only 4% in variance. It exposed the unreliability of the per-epoch estimator for model (A), where the impossible excursion below the Hartree-Fock limit is a direct consequence of $\text{Var}(E_L) = 3.21$. And it showed that the additional epochs in the longer (D+) run did not improve the wave function, despite a nominal energy gain.

The underlying reason is the zero-variance principle: energy and variance both vanish, relative to their exact limits, as Ψ_θ approaches an eigenstate, but they do so through different functionals of the error. An energy improvement unaccompanied by a variance improvement should be regarded with suspicion. We would argue that reporting the pair $(E, \text{Var}(E_L))$ should be routine rather than optional.

VI. LIMITATIONS AND OUTLOOK

Several limitations of this study should be stated.

The claims about nodal surfaces are inferred rather than measured. We observe that four ansätze sharing a determinant saturate at the same energy and that a nodal parameter breaks through it, which is consistent with the fixed-node argument but does not measure the fixed-node energy of either nodal structure. A diffusion Monte Carlo calculation on the two nodal surfaces would convert the inference into a measurement, and is the single most valuable extension of this work.

Model (A) is not a Hartree-Fock calculation. It is a three-parameter determinant optimised variationally, and it lies above the Hartree-Fock limit by 3.4 mHartree, as Table II shows. Where we quote the Hartree-Fock limit it is taken as an external reference value, not computed here.

The networks are small, with 16 hidden units and two layers. This was dictated by the requirement of several samples per parameter for the Stochastic Reconfiguration

metric rather than by expressive considerations, and the sample to parameter ratio of 10.7 sets a ceiling on the network size at fixed walker count. Whether a substantially larger network would change the conclusions is untested, although the saturation of the variance across models (B), (C) and (D) suggests that the limiting factor lies elsewhere.

Finally, the error bars depend on the decorrelation protocol. We found that increasing the separation between blocks from 30 to 150 sweeps changed the quoted uncertainty of model (D+) by roughly a factor of three, without changing the central value appreciably. Uncertainties in variational Monte Carlo are easily underestimated, and

the protocol used to obtain them should always be stated.

A further extension worth pursuing is zero-variance extrapolation. Since the energy error and the variance both vanish quadratically in the wave function error, a set of ansätze of differing quality should satisfy $E(\sigma^2) \approx E_0 + a\sigma^2$, and the intercept provides an estimate of E_0 independent of any single ansatz. The five models presented here span two orders of magnitude in variance and would supply the required points, although a controlled version of this analysis requires each model to be converged to its own variational minimum, a condition model (D+) does not yet meet.